

A Tandem Laboratory Scale Protein Purification Process Using Protein A Affinity and Anion Exchange Chromatography Operated in a Weak Partitioning Mode

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ABSTRACT: A significant consequence of scaling up production of high titer monoclonal antibody (mAb) processes in existing facilities is the generation of in-process pools that exceed the capacity of storage vessels. A semi-continuous downstream process where columns and filters are linked and operated in tandem would eliminate the need for intermediate holding tanks. This study is a bench-scale demonstration of the feasibility of a tandem process for the purification of mAbs employing an affinity Protein A capture step, followed by a flow-through anion-exchange (AEX) step with the possibility of adding an in-line virus filtration step (VF). All three steps were linked sequentially and operated as one continuous process using an ÄKTA FPLC equipped with two pumps and a system of valves and bypasses that allowed the components to be engaged at different stages of the process. The AEX column was operated in a weak partitioning (WP) mode enabled by a precise in-line titration of Protein A effluent. In order to avoid complex control schemes and facilitate validation, quality and robustness were built into the system through selection of buffers based on thermodynamic and empirical models. The tandem system utilized the simplest possible combination of valves, pumps, controls, and automation, so that it could easily be implemented in a clinical or commercial production facility. Linking the purification steps in a tandem process is expected to generate savings in time and production costs and also reduce the size of quality systems due to reduced documentation requirements, microbial sampling, and elimination of hold time validation.

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Introduction

Recent improvements in the upstream production of monoclonal antibodies (mAbs) and antibody-related proteins, with titers surpassing 5 g/L, have shifted the capacity bottleneck downstream, forcing manufacturers to consider increasing the productivity of the purification train (Shukla and Thömmes, 2010). A bold vision for purification of higher titer mAbs involves replacement of expensive and capacity-limited chromatography steps with column-free operations, such as precipitation (Low et al., 2007). In contrast, others argued for incremental improvement of the existing mAb purification platforms, which would continue to utilize existing manufacturing facilities without significant capital investments in retrofitting (Kelley, 2009). Downstream processes can be debottlenecked through integration of multiple unit operations into a single fluid system, thus eliminating in-process hold tanks and minimizing batch records and validation work. Such semi-continuous bioprocesses have been demonstrated. For example, perfusion cell culture was harvested by semi-continuous depth filtration, followed by an anion-exchange (AEX) membrane chromatography (Vogel et al., 2012). Another study reported continuous cell separation by alternating tangential flow (ATFTM) filtration linked to periodic counter-current (PCC) capture chromatography (Warikoo et al., 2012). Economic analysis of PCC predicted its viability for clinical, but not for commercial scale manufacturing of mAbs, due to substantial capital costs (Pollock et al., 2013). The tank volume limitation may be resolved by single pass tangential flow filtration

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(SPTFF) modules placed in-line between several process steps without introducing additional unit operations (Dizon-Maspat et al., 2012; Teske et al., 2010). The drawback of the SPTFF technology is that it does not completely eliminate the holding tanks, requires high membrane area, and has limited in-process control.

The present study addresses the potential purification bottleneck by demonstrating the feasibility of a semi-continuous tandem process at lab scale. The Protein A affinity step was linked to the AEX step followed by a virus filtration (VF) step via a continuous purification train within one cell culture harvest (Fig. 1). The bench-scale model utilized a single ÄKTA instrument equipped with standard components. Similar integration of multiple chromatographic steps into a single lab-scale unit operation has been

demonstrated in the drug discovery field, mainly for the purposes of increased throughput (Schlaeppli et al., 2006). The multi-dimensional preparative chromatography applications have been enabled by automation developed for ÄKTA Explorer instruments (Sigrell et al., 2003). However, these systems typically require an intermediate pool collection after the capture column for subsequent injection onto the polishing column and thus do not operate in a continuous mode. In contrast, the system described in this study integrates the Protein A capture column with a flow-through AEX column, while avoiding the intermediate pool collection by means of in-line titration. The choice of solutions obviated the need for pH feedback control during the titration. The feasibility study demonstrated that integration of the purification units would eliminate the need to store in-process material pools, reduce sampling, validation, and documentation requirements (e.g., bioburden sampling, batch record and documentation). The novel process allows for rapid and robust operation.

Materials and Methods

Chromatography Systems

All lab-scale runs were performed using an ÄKTA chromatography system obtained from GE Healthcare (Uppsala, Sweden), configured as shown in Figure 1b. The system contained two P-920 positive displacement pumps (A and B) capable of delivering a flow rate from 0.1 to 20 mL/min in the pressure range of 0–5 MPa. The accuracy of gradient composition was $\pm 2\%$ at 0.5–5 mL/min. The first pump directed the main flow and the second added titrant to the effluent from the Protein A column. During the operation of the tandem process, effluents from pumps A and B were directed into a M-925 mixer equipped with a 2-mL mixing chamber with a magnetic stirrer. Mixing was monitored in-line using a pH flow cell placed between the mixing chamber and Valve 2 as shown in Figure 1b. The three INV-907 valves were operated either to bypass the columns or to direct the solution into the columns. The UV and the conductivity signals were monitored in-line downstream of the AEX or the VF unit (when used). The pumps and data acquisition were controlled with UnicornTM 5.11 software (GE Healthcare). For each run, a pre-programmed method specifying the desired neutralization ratio was executed.

Antibodies

Two proprietary mAbs and one Fc fusion protein were employed to test the tandem purification model. These molecules have been previously well characterized, so that the in-process stability and targets for high-molecular weight (HMW) and other impurities were known. For each of these molecules, separate chromatography conditions had been previously developed for the Protein A and the AEX steps. This knowledge was utilized in designing the tandem processes. For example, specific Protein A wash buffer

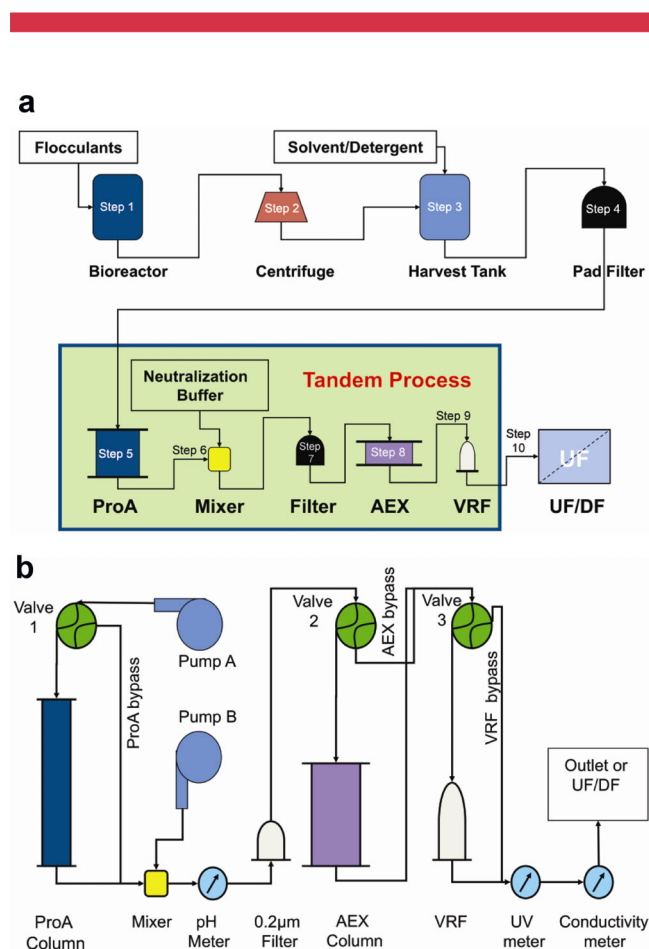


Figure 1. Generalized schematic representation of the tandem purification process: (a) as a part of an entire scale-up process for the manufacture of large amounts of protein (mAbs, Fc fusion proteins, etc.). *Pre-tandem process:* Flocculants are added to the bioreactor (step 1). The solution is then fed into the centrifuge to remove particles (step 2). Then the supernatant is fed to the harvest tank, where solvents and/or detergents are added for inactivation of enveloped viruses (step 3). Solids are removed using a pad filter (step 4). The feed is then run through the *tandem process* (tan box): Protein A affinity chromatography (step 5), mixer where the effluent is neutralized (step 6), AEX pre-column filter (step 7), AEX chromatography (step 8), virus reduction filter (VF, step 9), ultra-filtration/dia-filtration (UF/DF) where the protein is concentrated and exchanged into the formulation buffer and an additional low pH virus inactivation may occur (step 10); (b) at lab-scale with valves and pumps outlined.

Table I. An example of Protein A operation for mAb-2 for batch purification control.

Step	Buffer	Number of CV's
Equilibration	50 mM Tris, 0.15 M NaCl, pH 7.5	≥5
Load	Conditioned medium	Variable
Wash 1	20 mM Tris, 1.0 M Arginine, pH 7.5	≥5
Wash 2	10 mM Tris, 42 mM NaCl, pH 7.5	≥3
Elution	50 mM Glycine, 37 mM NaCl, pH 2.90	Variable
Post-elution wash	50 mM Glycine, 37 mM NaCl, pH 2.90	≥3
Neutralization	2 M Tris, pH 8.5	N/A
Strip	50 mM NaOH, 0.5 M Na ₂ SO ₄	≥5
Storage	16% EtOH, 0.15 M NaCl, 50 mM Tris pH 7.5	≥5

(Wash 1, Table I) for mAb-2 was adapted from the development program. Similarly, loading and washing conditions for the AEX step were also known, which enabled models for optimization of Protein A elution and neutralization buffers and the target neutralization ratio.

Chromatography Columns and Filters

The Protein A capture step was accomplished using MabSelect affinity medium obtained from GE Healthcare. The Protein A resin was packed into columns with bed heights of 20–25 cm and an internal column diameter of 1.1 cm (Vantage VL 11 × 250, EMD Millipore, Billerica, MA). The step was operated in a range of linear velocities from 100 cm/h to 300 cm/h. An example of Protein A operation for mAb-2 for batch purification is shown in Table I. A tandem purification of mAb-2 employed the same conditions, except the elution buffer was changed to 50 mM Gly, 17 mM NaCl, 20 mM Tris, pH 3.7 and the neutralization buffer was 0.2 M Tris, 0.25 M NaCl, pH 9.5.

The AEX chromatography employed Fractogel TMAE HiCap (M; EMD Millipore), a strong anion exchange resin. The resin was packed into Vantage columns of 1.6 cm i.d. × 10–15 cm bed height and run at a linear velocity of 50–150 cm/h. The cross-sectional area of the Protein A column was half of that of the AEX column, so that at a maximum Pump A flow rate of 5 mL/min, the linear velocities through the columns were 300 and 150 cm/h, respectively. An example of AEX operation for mAb-2 for batch purification is shown in Table II. During the tandem operation, the AEX “Equilibration 2” buffer was replaced with the Protein A “Wash 2” buffer (Table I) titrated in-line with the neutralization buffer in a pre-determined ratio. Similarly, the AEX load was the neutralized Protein A elution, while the “Wash” was the neutralized Protein A “Post-elution wash”. The VF filter employed in the studies was Millipore Viresolve Pro 3.1 cm² with dual-layer PES membrane.

Determination of the k_p

Weak partitioning AEX (WP-AEX) chromatography requires a precisely formulated mobile phase to promote retention of

Table II. An example of AEX operation for mAb-2 for batch purification control.

Step	Buffer	Number of CV's
Equilibration 1	50 mM Tris, 2.0 M NaCl, pH 7.5	4
Equilibration 2	50 mM glycine, 42 mM NaCl, 20 mM Tris, pH 7.8	≥2.5
Load	Neutralized protein A peak	Variable
Wash	50 mM glycine, 42 mM NaCl, 20 mM Tris, pH 7.8	3
Strip	50 mM Tris, 2.0 M NaCl, pH 7.5	N/A
Regeneration	500 mM NaOH, 0.5 M NaCl	≥5
Storage	16% EtOH, 0.15 M NaCl, 50 mM Tris, pH 7.5	≥5

stronger-binding impurities, such as HMW species and host cell proteins (HCP) at the expense of a weaker-binding product (Kelley et al., 2008). The binding strength is determined by a thermodynamic partition coefficient, k_p , which is defined as the ratio of adsorbed protein concentration (Q) to free protein concentrations (C) at equilibrium according to Equation (1).

$$k_p = \frac{Q}{C} = \frac{\text{bound}}{\text{free}} \quad (1)$$

Protein binding to an AEX resin is a function of pH and concentration of anions (typically, $[\text{Cl}^-]$) in the AEX equilibration, loading, and wash buffers. More broadly, the anion concentration is proportional to ionic strength, φ , which will be used in the calculations. During process development, the WP AEX operating conditions are determined in high-throughput batch-binding experiments. For each molecule, the function (f) in $k_p = f(\text{pH}, \varphi)$ is empirically derived by collecting values of protein concentration after incubation with the AEX resin at various pH and φ conditions. The k_p for each condition is calculated according to (1) and, using JMP software (SAS, Cary, NC), fitted to a polynomial of a general form (2):

$$k_p = a \times \text{pH} + b \times \varphi + c \times \text{pH} \times \varphi + d \times \text{pH}^2 + e \times \varphi^2 + f \quad (2)$$

where φ is ionic strength, and a, b, c, d, e , and f are empirical constants providing the best fit for the experimental k_p data.

In order to set the target volumetric ratio of flow rates from pumps A and B, (neutralization ratio, ν), a numerical model was created in an Excel spreadsheet for each molecule. First, the pH of the neutralized Protein A elution for a given ν was calculated based on Henderson–Hasselbalch equation for the Tris buffer system. A mass balance provided the final chloride concentration φ for each ν . The resulting pH and φ functions of ν were then substituted into Equation (2) and plotted. The target ν was chosen based on desired k_p (Kelley et al., 2008) or in the middle of the robust range of k_p with respect to ν .

Flocculation and Detergent Treatment

Co-precipitation of cells from the cell culture fluid with calcium phosphate (flocculation) followed by pad filtration (using a depth filter containing cellulose and inorganic filter aid) has been shown to significantly reduce turbidity in the neutralized Protein A peak (Coffman J. Shpritzer, R. and Vicik, S., International Conference on Recovery of Biological Products XII, 2006; Coffman et al., Patent no. US 7,855,280). In a typical process, the non-flocculated pad filtered conditioned media (CM) is held in a harvest tank before loading on a Protein A column. The current study utilized CM harvested by this method and subsequently frozen at -80°C . Therefore, the order of operations in this study was different than that proposed in Figure 1 for the tandem process. After thawing, the CM was flocculated by mixing with 30 mM CaCl_2 , 20 mM K_2HPO_4 , 40 mM HEPES, pH 7.5 at room temperature for 1 h followed by centrifugation in a benchtop centrifuge at 340 g for 10 min to remove formed solids. In some experiments, flocculated CM was spiked with 0.5% (w/v) or 0.25% (w/v) Triton X-100 detergent. The resulting pool was then filtered through a $0.2\text{ }\mu\text{m}$ PES filter.

Analytics

In addition to UnicornTM-based data acquisition, separate pressure traces were obtained using SciPres in-line pressure sensors connected to SciPres pressure monitor box (SciLog Bioprocessing Systems, Inc., Madison, WI). Protein concentration was measured using Varian UV-Vis spectrophotometer at 280 nm. Turbidity was measured by a hand-held Hach turbidimeter (Hach Company, Loveland, CO). Percent HMW was determined by size exclusion high-performance liquid chromatography (SE-HPLC) using TSK G3000 SW_{XL} columns (Tosoh Bioscience, King of Prussia, PA) connected to Waters Alliance 2795 Separations Module (Waters

Corporation, Milford, MA). Concentration of Triton X-100 in the tandem peak fractions was determined by reversed-phase (RP) HPLC, using a C_{18} column. Residual Protein A and HCP levels were measured using proprietary ELISA protocols.

Results and Discussion

Flocculation of the Conditioned Media

Flocculation of CM via addition of calcium phosphate or a charged polymer had been shown to promote the efficiency of settling, clarification, filtration and centrifugation operations by removing the cell debris (Roush and Lu, 2008). Incidentally, flocculation also removes HCP and reduces the level of precipitation during the Protein A peak neutralization and prevents fouling of the AEX column, as demonstrated in earlier work (Coffman et al., Patent no. US 7,855,280). Filtration of CM through a pad filter containing diatomaceous earth, followed by flocculation, significantly reduced turbidity in the Protein A peak apex. Although turbidity by itself is not an accurate predictor of filter plugging, it serves as a convenient proxy measurement for the concentration of solids. The experiment shown in Figure 2 demonstrated that the turbidity of the neutralized Protein A fractions of mAb-1 immediately after the peak apex were 32 NTU when flocculated, compared to 227 NTU without flocculation. Based on this result, all tandem experiments described in this paper were generated with flocculated CM. None of the tandem experiments showed any sign of pre-AEX filter plugging or AEX column fouling, suggesting that precipitation of Protein A peak could be successfully mitigated and an in-line chromatography step could be used after the Protein A column. Follow-up experiments will need to be performed with the order of operations outlined in Figure 1, so that the

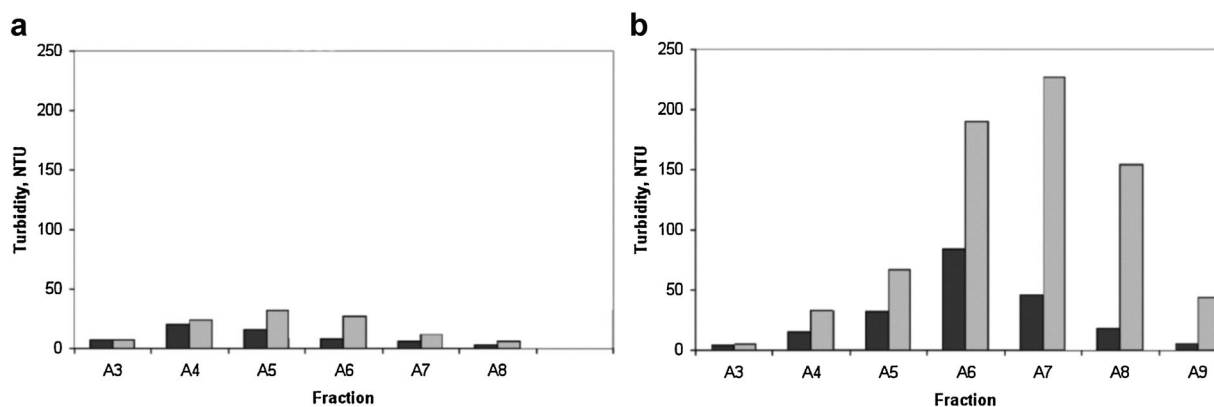


Figure 2. Turbidity of the Protein A peak fractions (approximately 0.2 CV each) of mAb-1, pre- and post-neutralization. **a:** Flocculated and pad-filtered load. **b:** Pad filtered load, without flocculation. Fraction A5 corresponds to the peak apex, with protein concentration of 55 mg/mL. Color code: pre-neutralization turbidity (dark gray); post-neutralization turbidity (light gray).

pad filtration follows the flocculation step. However, the overall trend in turbidity reduction of the Protein A peak is expected to be maintained.

Viral Clearance

Downstream processing of biotherapeutics typically includes several virus inactivation or removal steps (Miesegaes et al., 2010). The elimination of a low-pH hold after the Protein A column, which is a classic way to inactivate enveloped viruses (Brorson et al., 2003), meant that an alternative virus inactivation step had to be considered for the tandem process. In addition, viral clearance over the tandem columns would be difficult to validate due to variable pH, ionic strength, and protein concentration. Therefore, viral clearance through two tandem chromatography steps will not be claimed in early phases of clinical production. The entire Protein A-AEX assembly may be treated as a disposable plug-in device, not validated for virus removal. Typically, AEX contributes $>4 \log_{10}$ reduction value (LRV) of murine leukemia virus (MuLV) as reported by Kelley et al. (2008). In order to compensate for the loss of the AEX step for MuLV removal, a detergent inactivation step was introduced in the proposed process (Fig. 1, Step 3). Incubation of an in-process pool of murine mAb with 0.5% Triton X-100, a non-ionic detergent, has been shown to provide ≥ 3.8 LRV of MuLV at 4°C over the course of 4 h (Hilfenhaus et al., 1991). More recent studies have demonstrated robust inactivation of xenotropic murine leukemia virus (XMuLV) by $\geq 0.2\%$ of Triton X-100 in clarified conditioned media (Quertinmont M, Meeting presentation, IBC 2011; Nolting J et al., Abstract BIOT 56, ACS National Meeting, 2008). These studies found ≥ 5 LRV of clearance in a wide range of protein concentration and temperature (15–25°C) during 60 min incubation. In the current study, the detergent inactivation step was combined with flocculation of clarified CM. Triton X-100 was spiked into CM to a concentration of 0.5% (w/v). The RP-HPLC analysis of the fractionated Protein A peak demonstrated that the detergent was largely removed across the Protein A step from 0.5% (w/v) in the load to 0.0135% (w/v) in the peak apex, which is a $1.5 \log_{10}$ reduction.

An additional low pH inactivation could be implemented during the UF/DF (Fig. 1, Step 10). Intermediate buffer exchange into a buffer with $\text{pH} \leq 3.8$ has been shown to achieve ≥ 4.6 LRV of XMuLV (Brorson et al., 2003). Thus, a combination of detergent treatment (≥ 5 LRV), VF (4 LRV, as reported by Marques et al., 2009) and low-pH inactivation (≥ 4.6 LRV) should achieve approximately 14 LRV of enveloped virus and >5 LRV of parvovirus (Hongo-Hirasaki et al., 2010), without even taking into account virus removal over the tandem columns. Such level of clearance should be acceptable for the early clinical filings in the US. For later stage clinical campaigns, additional studies may be performed in order to validate viral clearance over the tandem step by bracketing the most unfavorable conditions (such as under-titrated protein A effluent, high protein concentration, or high level of HMW).

Effect of Protein Concentration Spike on WP-AEX and VF Performance

The total protein challenge affects the quality of the WP-AEX pool, as more impurities are leached from the column as more product weakly binds to it. The effect of a product concentration spike on WP-AEX chromatography was demonstrated using Protein A peak of mAb-1. In a control experiment (Fig. 3a), the Protein A pool was mixed, neutralized, and loaded onto the AEX column. The same process was repeated with fractionated Protein A peak, where each fraction was neutralized separately and sequentially loaded onto the AEX at a continuous flow rate (Fig. 3c). Both conditions provided similar recoveries and HMW removal, and displayed similar UV-chromatograms. More importantly, the multimeric aggregates (HMW1) did not break through the AEX column during the passing of the peak apex. The breakthrough of the dimer species (HMW2) was expected. This proof-of-principle experiment showed that a Protein A peak apex with a protein concentration of approximately 55 mg/mL could be directly loaded onto the AEX column without causing breakthrough of HMW1 and other impurities.

Purification of the mAb-1 was then repeated in a tandem mode, with columns and sensors linked together as shown in Figure 1b. The off-line analyses of fractions are shown in Figure 4. The gradual rise in transmembrane pressure over the VF step during the tandem elution of mAb-1 correlated with the protein concentration spike. However, after the elution, the pressure returned to almost the pre-elution level, suggesting that the pressure rise was due to viscosity of the product peak and not due to plugging of the membrane. At the AEX peak protein concentration of approximately 30 mg/mL (the Protein A peak apex of 55 mg/mL spread out during the passage through the AEX column), the observed pressure drop across the VF was approximately 25 psi. The total HMW level in the tandem fractions was similar to that observed in the pooled run (Fig. 3b). The protein concentration spike will likely have a minimal effect on LRV of the VF step, since it has been found that both XMuLV and murine minute virus (MMV) were removed at 3–4 LRV by Planova™ 20N filter independent of mAb concentration in the range of 4–14 mg/mL (Marques et al., 2009). Another study reported a consistent LRV > 5 of porcine parvo virus (PPV) by Planova™ 20N filter at IgG concentrations 1–50 mg/mL (Hongo-Hirasaki et al., 2010).

Effect of Variable pH on WP-AEX and VF Performance

As the AEX step is operated in a WP mode, both the pH and conductivity of the solutions at this step are important to regulate, as this maintains both yield and purity of the resulting antibody. However, running the tandem at a constant neutralization ratio inevitably leads to a pH drift of the AEX load from more basic levels at the beginning of the peak to more acidic levels toward the end, as the Protein A effluent is mixed with the high-pH titrant. Furthermore, both

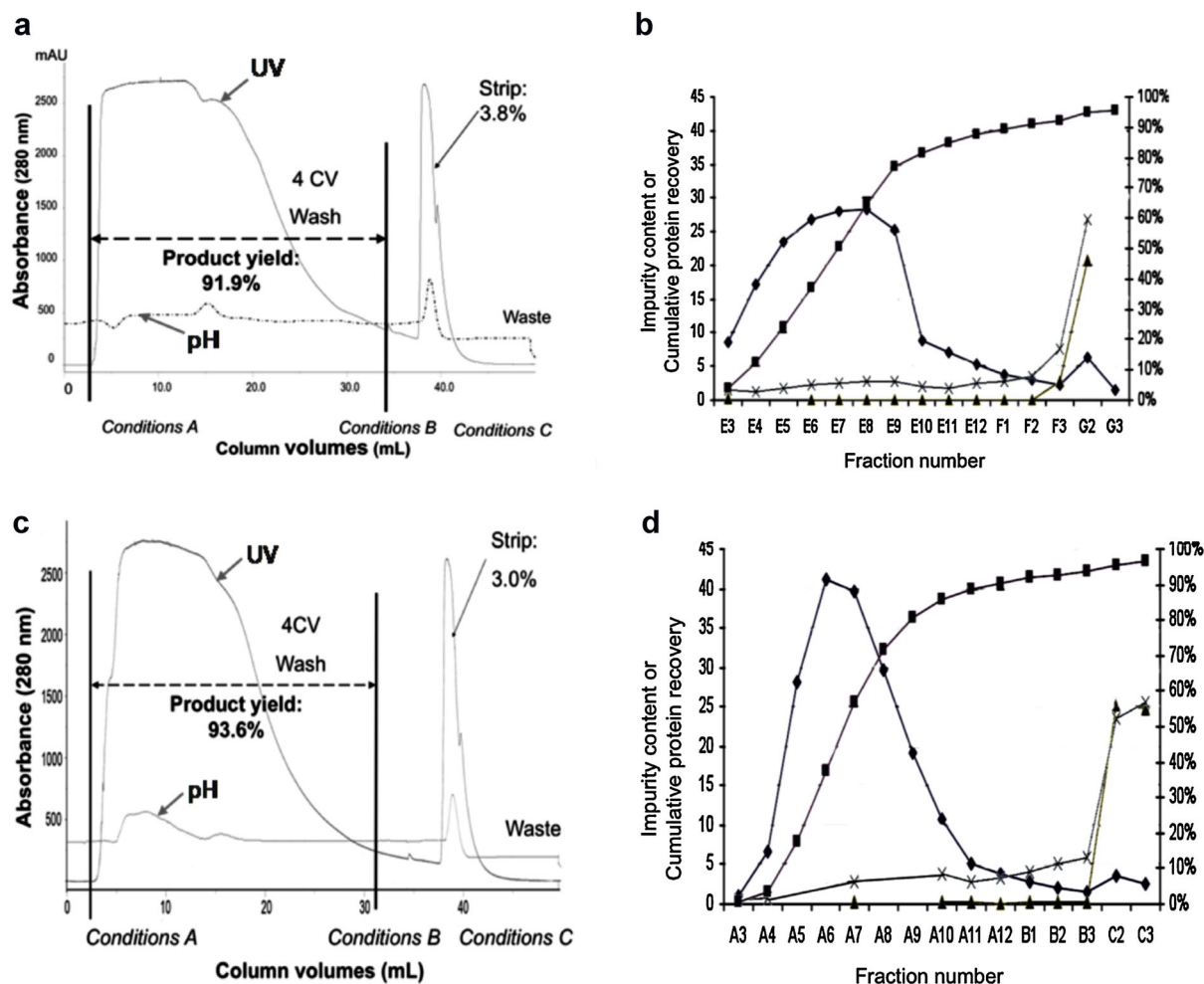


Figure 3. AEX chromatograms and corresponding off-line measurements of impurity content or product recovery of mAb-1. **a:** Chromatogram resulting from loading of pooled neutralized Protein A peak; **b:** off-line measurements in AEX fractions from the pooled loading; **c:** Chromatogram resulting from sequential loading of neutralized Protein A peak fractions from a similar Protein A run; **d:** off-line measurements in AEX fractions from the sequential loading. *Wash Buffer:* 20 mM Tris, 32 mM NaCl at pH 7.78; *Strip buffer:* 50 mM Tris, 2 M NaCl at pH 7.5. Total protein concentration (mg/mL, ◆), % HMW1 (multimer, ▲), % HMW2 (dimer, ×), cumulative protein recovery (■).

the Protein A ligands and the adsorbed product have residues that can be protonated, thus taking up acid in the elution buffer. These phenomena explain the observed shape of the pH trace recorded after the mixer, which exhibits a pH swing. In the example shown in Figure 4, an off-line buffer titration experiment correctly predicted that the final pH of the titrated elution buffer would stabilize at $\sim$ pH 7.9. However, the early fractions of the elution buffer were nearly devoid of acid and, as a consequence, became over-titrated with the neutralization buffer. The resulting pH excursion to pH 8.3 occurred during the peak apex. This change in pH can cause an increase in product binding to the AEX column operating in a WP mode, thereby negatively affecting the product recovery. However, at the end of the Protein A elution, the pH reached the target level causing the desorption of all extra bound material. The observed levels of HMW in the post-VF fractions were still within limits. The

HCP level was reduced to less than 10 ppm, versus 72,000 ppm of HCP present at the start of the process, and 55,000 ppm after the flocculation.

The effects of wide pH and ionic strength variations on VF flux have been previously reported (Hongo-Hirasaki et al., 2010). Predictably, a decay in VF flux was observed at very low ionic strength (<10 mM) and at the pH approaching the *pI* of the mAb. However, such an effect during the tandem processing would be minimal due to comparatively small actual fluctuations of pH and ionic strength. One would not expect a significant decrease in virus LRV due to these fluctuations.

High Impurity Challenge

An example of HMW1 breakthrough during a tandem purification of an Fc fusion protein is shown in Figure 5. This

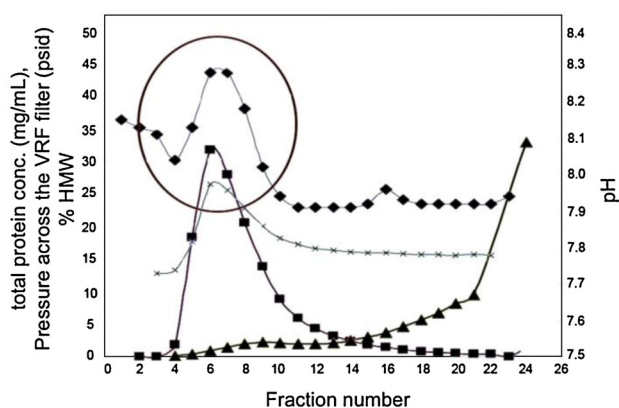


Figure 4. Off-line determination of post-Protein A pH, protein concentration (mg/mL), VF trans-membrane pressure (psid), and % total HMW in the fractions collected after the tandem purification of mAb-1. Circled area points to the excursion in the pH of the AEX load. Code: pH (◆), % total HMW (▲), Cross-VF pressure (psid, ×), protein concentration (mg/mL, ■).

product typically has very high levels of aggregation in the CM, so that the Protein A pool usually contains 15% of HMW1 and 15% of HMW2. Loading of the pooled Protein A peak on the AEX column (control run, Fig. 5a) resulted in an almost immediate breakthrough of weaker binding HMW2 and a slow appearance of HMW1 at the tail end of the chromatogram. In a tandem process (Fig. 5b), HMW1 broke through shortly after the peak apex. The observed HMW1 breakthrough was probably due to a temporary saturation of binding capacity of the AEX resin as a wave of impurities moved through the column. Modification of Protein A elution buffer, as described in Figure 5 caption, spread out the Protein A elution peak and stabilized the pH, which prevented excessive binding of product to the AEX resin and improved removal of HMW1 (Fig. 5c). This experiment illustrated some limitations of the tandem process when applied to products with very high levels of impurities.

Tandem Operation Robustness With Respect to Neutralization Ratio

The key to successful application of the tandem process is to run the AEX column for all purifications in WP mode within a k_p range of 0.3–3 (Kelley et al., 2008). Theoretically, the target k_p could be maintained through a feedback mechanism based on pH and conductivity signals after the mixer. Alternatively, the volume fraction of the titrant, or neutralization ratio (v), can be held constant at a pre-determined level throughout the run. One may anticipate, however, that the success of this approach largely depends on the ability of the pumps A and B (Fig. 1b) to deliver a constant and precise ratio of titrant to the product stream. While the maintenance of a constant v is generally not a problem using an ÄKTA FPLC, it may become an issue at scale, where a pump flow rate is not as accurate and depends on back pressure.

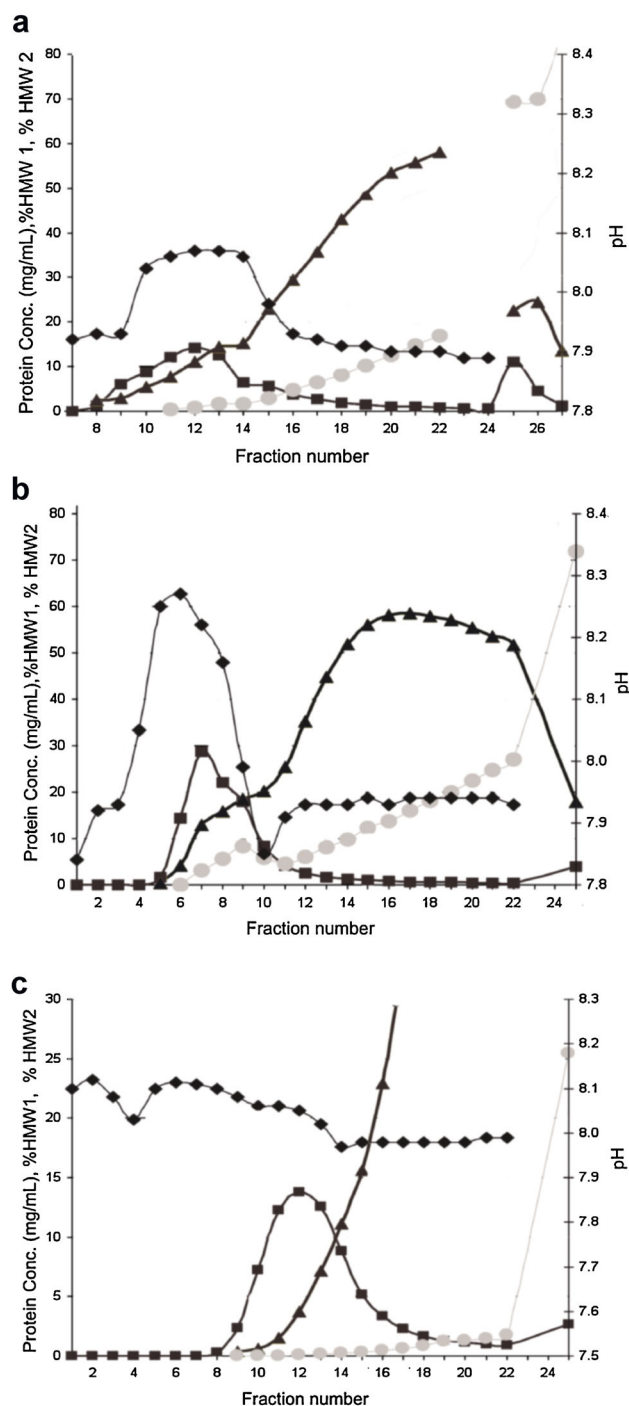


Figure 5. Off-line determination of pH, protein concentration (mg/mL), and % HMW species in the fractions collected during purification of the Fc fusion protein at $k_p = 1.5$. **a:** non-tandem AEX run with pooled neutralized Protein A peak as a load (control); **b:** tandem Protein A-AEX, with aggressive elution (100 mM Gly, pH 3.5); **c:** tandem Protein A-AEX, with mild elution (80 mM Gly, 20 mM Tris-HCl, pH 3.75). The addition of Tris-HCl to the elution buffer mitigated against pH fluctuations, while a higher pH elution buffer resulted in a more spread-out elution peak, which reduced the breakthrough of HMW1. Code: pH (◆), % HMW2 (dimer, ▲), % HMW1 (multimer, ●), total protein concentration (mg/mL, ■).

In this study, k_p was controlled through a self-correcting buffer system, in which a constant k_p was maintained in a wide range of v . The key feature of this system is the high ionic strength of the high-pH titrant. In case of over-titration, the pH of the AEX load increases, but so does its ionic strength resulting in a small net effect on the k_p . A similar result is expected in the case of under-titration, whereby a lower than target pH will be compensated by a lower ionic strength of the AEX load, again resulting in the maintenance of a constant k_p . Therefore, with a correct choice of Protein A elution and neutralization buffers a rather wide process robustness window is possible.

The robustness model was created for mAb-2. The Standard mode of neutralization (Fig. 6a) was based on the batch process conditions developed for the Protein A step (Table I) and AEX step (Table II). The k_p versus v curve was then modified by changing the input parameters: pH, concentrations of the buffering species (Glycine and Tris), and salt concentration in both buffers. The iterations were repeated until a “flat” k_p versus v profile was obtained (Robust mode, Fig. 6a). The model predicted that in order to flatten the k_p versus v profile, the Protein A elution buffer needs to have higher pH and lower salt concentration than in the Standard mode, while the neutralization buffer should have higher pH and higher salt.

The model was validated by performing the tandem purification runs on mAb-2 in Robust and Standard modes using the three levels of v : 6.5, 11.5, and 16.5% (v/v). In this case, the center point $v = 11.5\%$ (v/v) was predicted to result in the same target k_p value of 1.25 in both modes of operation. However, as seen in Figure 6a, over-titration and under-titration by 5% (v/v) were expected to produce dramatically different outcomes. Figure 6b shows the UV absorbance chromatograms for all six cases. As predicted, the elution profile of the AEX column changed considerably in the Standard neutralization mode, with the under-titrated run showing the earliest breakthrough of product, while the over-titrated run resulted in significant product binding and slow elution. In the robust mode the elution profiles appeared very close at all three levels of v . Analysis of the tandem mAb-2 pools from the runs showed that the Standard/under-titrated run resulted in high impurity levels (3.3% HMW1, 30 ppm of HCP and 4.3 ppm of leached protein A), while the Robust/under-titrated run produced a pool with 0.14% HMW1 and other impurities at BLOQ levels. Product recoveries from these runs were not recorded. However, based on the chromatogram shape of the Standard/over-titrated run, significant product losses could be expected if the peak collection was limited to a certain number of column volumes. The results from the mAb-2 tandem runs validated the model of built-in process robustness.

Conclusions

In this study, the feasibility of a semi-continuous tandem process containing a Protein A affinity column, an anion exchange column, and a virus reduction filter was demon-

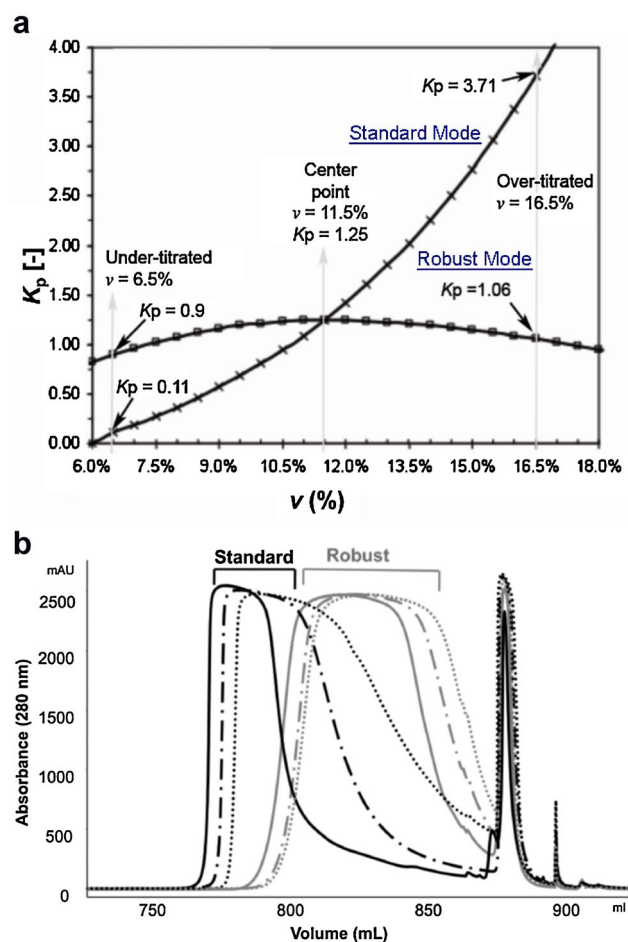


Figure 6. Simulations of the k_p as a function of v for mAb-2 using the standard neutralization and the robust neutralization conditions. **a:** k_p as a function of v under a standard neutralization mode (-x-) and robust mode (-□-). The standard mode is based on conditions developed for batch processes as summarized in Table I and Table II. In the Robust mode the Protein A elution buffer was changed to 50 mM Gly, 17 mM NaCl, 20 mM Tris, pH 3.7 and neutralization buffer was 0.2 M Tris, 0.25 M NaCl, pH 9.5. **b:** Corresponding tandem (Protein A-AEX) process chromatograms showing UV traces obtained in the standard neutralization and robust neutralization modes at different v . Standard neutralization (black lines), robust neutralization (gray lines) was performed at $v = 6.5\%$ (solid line, —), $v = 11.5\%$ (dashed line, — · —), and $v = 16.5\%$ (dotted line,).

strated. Two different mAbs and one Fc fusion protein were tested. Product quality (HCP, leached protein A, high molecular weight aggregates, and antibody recovery) in the tandem pools was comparable to that of standard purification methods. Precipitation of impurities during Protein A peak neutralization was avoided by flocculation of conditioned media with calcium phosphate. In most cases, fluctuations in pH and protein concentration did not have a significant effect on the performance of the AEX and VF steps. Models based on thermodynamics and empirical HTS results were used to optimize the compositions of Protein A elution and neutralization buffers in order to ensure robust AEX performance with respect to the neutralization ratio. Thus, quality was designed into the process to avoid complex

control systems. Not only does the tandem process eliminate the need for large tanks to handle the in-process pools, the overall process can be run faster than a traditional non-tandem process, since many of the wash steps are shared between the columns. Further, automation of the entire system reduces the number of batch record entries and operating costs. The elimination of in-process pools also reduces the number of samples to test for bio-burden and endotoxin and minimizes the possibility of external contamination.

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